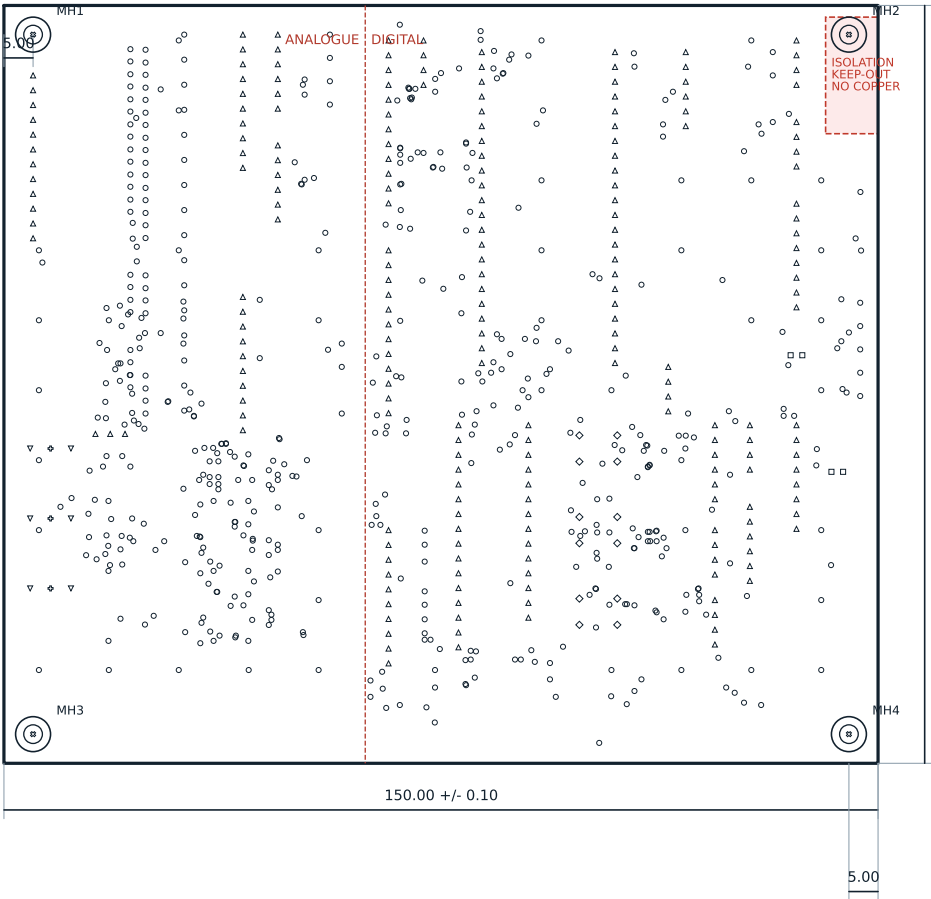




DRILL SCHEDULE (finished hole sizes)

sym	dia mm	qty	plated	used for
○	0.30	552	PTH	through vias
□	0.90	4	PTH	J13 / J24 JST PH
△	1.00	217	PTH	2.54 mm socket strips
◇	1.20	12	PTH	SW1-SW3 tactile switches
▽	1.50	6	NPTH	DIN 42802 retention posts (NPTH)
⬢	1.70	3	PTH	J15-J17 DIN 42802 signal pin
⊗	3.20	4	NPTH	M3 mounting holes (NPTH)

798 holes in total. Tolerance +/-0.08 mm on plated, +/-0.05 mm on non-plated.

FABRICATION NOTES

- Board outline 150.0 x 130.0 mm, rectangular, no cut-outs, no slots.
- FOUR layers, FR-4, Tg >= 150 C, 1.60 mm +/- 10 % finished.
L1 signal, L2 reference plane, L3 reference plane, L4 signal.
Outer copper 1 oz (35 um) finished; inner copper 0.5 oz (17 um).
Stack: 0.035 / prepreg 0.2 / 0.017 / core 1.065 / 0.017 / prepreg 0.2 / 0.035.
- Surface finish ENIG: Au 0.05-0.10 um over Ni 3.0-6.0 um.
- Solder mask green LPI both sides; silkscreen white, both sides.
- Minimum track 0.20 mm, minimum clearance 0.20 mm (8 mil / 8 mil). Most conductors are 0.25 mm or wider; the exceptions are listed in the DRC report.
- Vias 0.60 mm pad / 0.30 mm finished hole, tented on both sides.
- Smallest plated hole 0.30 mm (vias); largest 1.70 mm (J15-J17).
- Four non-plated 3.2 mm holes at (5,5) (145,5) (5,125) (145,125); keep 6.0 mm clear of copper on both layers.
- IPC-6012 class 2, IPC-A-600 class 2. 100 % electrical test to the supplied IPC-D-356A netlist.
- No controlled impedance is required. USB_DP and USB_DN are a 0.30 mm pair on 0.35 mm spacing over the bottom-layer DGND pour: about 95 ohm differential on this stack-up. No IMPEDANCE coupon and no impedance report are required. A microsection coupon IS required -- see note 13.
- Mark the board with the fabricator's date code and UL mark inside the outline on the bottom silkscreen, clear of all pads.
- Panelisation is the fabricator's choice. V-score or tab-route, 5 mm rails.
- LOT COUPON. Include ONE four-layer microsection coupon per fabrication lot, in the panel rails, carrying at least one plated through-hole of each drill size and one via. It is sectioned at incoming inspection to verify layer order and internal annular ring. OP-EEG-010 row IQC-B11 REJECTS THE WHOLE LOT on wrong layer order or an internal annular ring below 0.025 mm, so a lot that arrives without a coupon cannot be accepted at all.
- LOT DOCUMENTS. Supply with every lot, referenced to the lot number:
(a) certificate of conformance naming this drawing and revision (IQC-B1);
(b) 100 % electrical test report against the supplied IPC-D-356A (IQC-B2);
(c) ENIG thickness report, XRF, gold and nickel, min 3 points (IQC-B5);
(d) layer-to-layer registration report (IQC-B12);
(e) the microsection report from the note 13 coupon (IQC-B11).
These are conditions of acceptance, not extras. Each is an incoming-inspection row in QP-EEG-010 section 2.1 and each rejects the lot if it is absent.

STACK-UP

solder mask, LPI green
L1: F.C.U. signal, 1 oz (35 um) + ENIG
prepreg 0.200 mm
L2: In1.Cu reference plane, 0.5 oz (17 um)

core 1.065 mm, FR-4 Tg >= 150 C

L3: In2.Cu reference plane, 0.5 oz (17 um)
prepreg 0.200 mm
L4: B.Cu signal, 1 oz (35 um) + ENIG
solder mask, LPI green

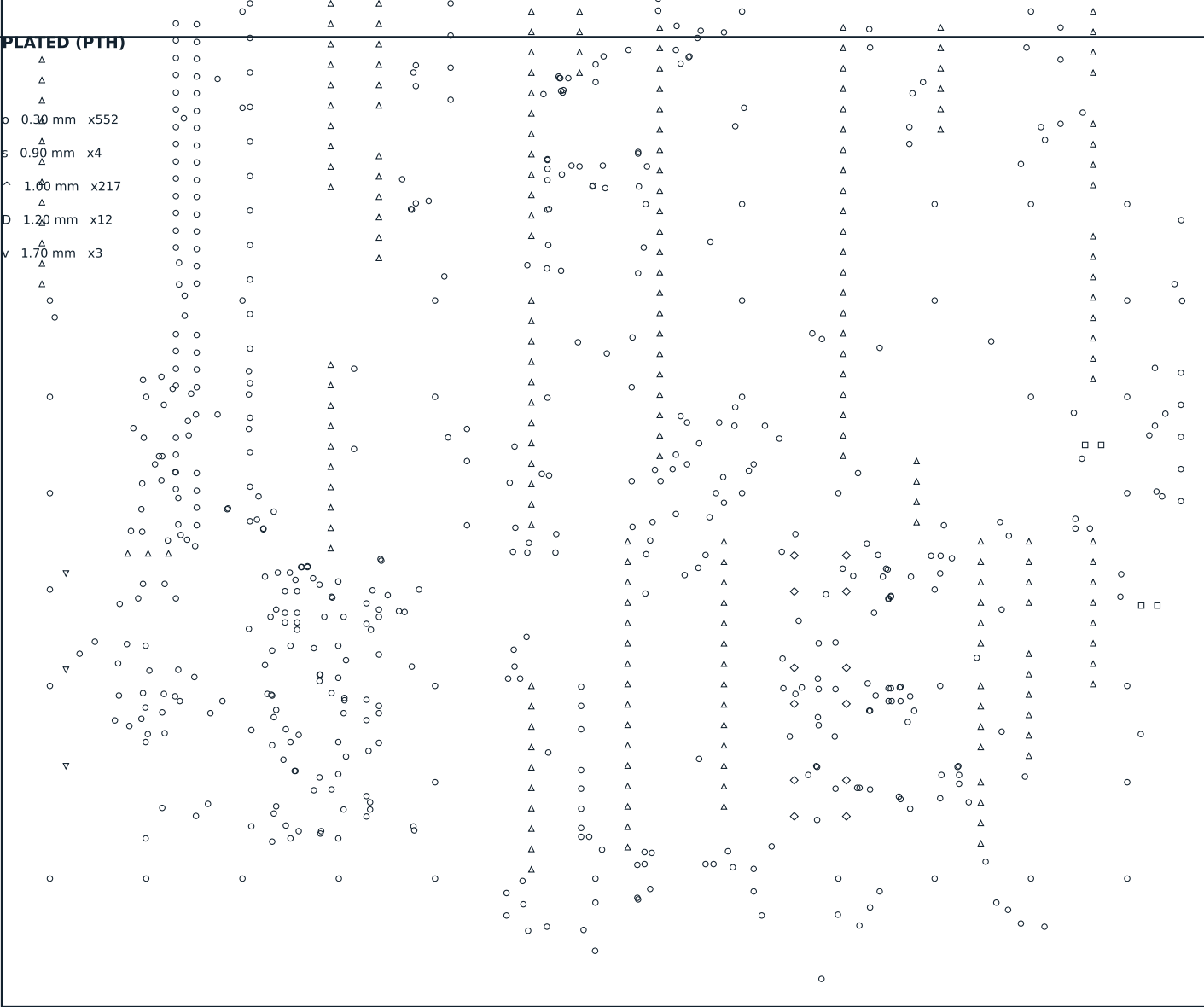
Total 1.60 mm +/- 10 %. Symmetric stack, so the board should not bow. Through vias only: no blind or buried vias, no back-drill, no via-in-pad, no filled or plugged vias.

EEG-CAR-01 -- fabrication drawing

4 layers 150.0 x 130.0 mm, 1.60 mm FR-4, ENIG. Scale 1:1 on A3.
IT One Voice research programme -- CC BY-SA 4.0

PLATED (PTH)

- 0.30 mm x552
- 0.90 mm x4
- 1.00 mm x217
- 1.20 mm x12
- 1.70 mm x3



NON-PLATED (NPTH)

- 1.50 mm x6
- 3.20 mm x4



NON-PLATED HOLES CARRY NO COPPER AND NO SOLDER-MASK OPENING.
The four 3.2 mm holes are M3 mounting holes with a 6.0 mm diameter copper keep-out on both layers.
The six 1.5 mm holes are the retention posts of the DIN 42802 sockets J15-J17 and need only a 1.10 mm copper gap.
Both sets are supplied in EEG-CAR-01-NPTH.drl and must NOT be merged into the plated file.

EEG-CAR-01 -- drill map and hole detail

Plated and non-plated holes shown separately